Hackathon 2026

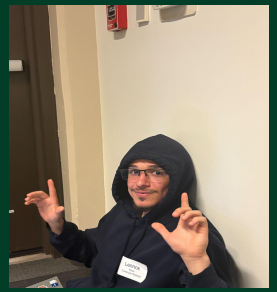
Team #17

TEAM #17 Members

Saren Sanjai
Freshman

Rio Johnson
Junior

Leia Dove
Junior



Lennox Nutall
Senior

PROMPT

#1

Objective

Predict corn grain yield for 5 states across 4 forecast periods, including uncertainty ranges.

#2

Forecast Periods

- August 1
- September 1
- October 1
- Final – End of season

#3

States

Iowa, Colorado, Wisconsin, Missouri, Nebraska

#4

Output

Yield prediction + cone of uncertainty for each state and forecast date

Data

USDA NASS

- **What**: Country-level corn grain yield results in BU/ACRE
- **Why**: records used as target variables that model trains to predict

NAIP

- **What**: 1-meter spatial resolution tiles (4 bands)
- **Why**: helps model distinguish between high-performing and stressed zones within same HLS pixel

Weather

- **What**: Weather conditions (i.e., precipitation, mean/min/max temp, etc)
- **Why**: GDD informs model of crop stage based on conditions & plant date
- An all-hands meeting

CDL

- **What**: Annual 30m rasterland cover classification
- **Why**: ensures model ignores noise from other crops like soybeans

HLS S30

- **What**: 6 Spectral bands
- **Why**: Primary input for model because Prithvi is pre-trained on HLS data

INVESTIGATION

IDE

Jupyter Notebook – Best for data exploration, testing pipelines, visualizations, and step-by-step model experiments.

VS Code – Best for structured development, managing multiple files, debugging scripts, version control, and production-ready code.

AWS

Used *S3 Buckets* for dataset storage, *SageMaker* for model development/training, *Amazon Bedrock* to explore LLM tools, and *Amazon Q / Quick AI* for coding and workflow assistance.

PRITHVI

NASA/IBM geospatial foundation model used to analyze satellite imagery and generate features for crop yield forecasting.

Initial Approach

- Integrate a nitrogen dynamics sub-model with Prithvi to fuse process-based agronomy with satellite-driven feature extraction
- Use the nitrogen sub-model to estimate plant-available nitrogen from weather, soil, and moisture inputs
- Provide nitrogen-derived features as additional input alongside satellite imagery
- Feed nitrogen-informed features into Prithvi to help contextualize vegetation patterns
- Train the integrated system end-to-end to improve prediction accuracy, robustness, and generalization across varying environmental conditions

PIPELINE

Connect to AWS S3 Bucket

Our notebook reads HLS satellite data stored in our S3 bucket (corn-yield-pipeline) for each target state.

Select Satellite Bands

The code searches each state folder for:

1) B04 = Red band
2) B8A = Near Infrared band

Calculate NDVI

Using Red + NIR values, the notebook computes NDVI to measure vegetation health and crop strength.

Estimate Base Yield by State

Each state starts with a baseline corn yield, then adjusts that number using the NDVI score.

PIPELINE

Forecast 4 Time Periods

The code generates yield estimates for:

- Aug 1
- Sept 1
- Oct 1
- Final / Post-Harvest

Create Uncertainty Cone

Each forecast gets a low and high range using preset uncertainty percentages that shrink later in the season.

Output Results

Final predictions are saved into json with:

1) Yield estimate
2) Low / High range
3) Mean NDVI

Results

Predicted growth forecasted by our model by study state

Iowa

Aug 1: 166.0 +/- 19.9 bu/acre

Sep 1: 175.0 +/- 15.8 bu/acre

Oct 1: 180.4 +/- 10.8 bu/acre

Post Harvest: 184.1 +/- 7.4 bu/acre

Wisconsin

Aug 1: 157.2 +/- 18.9 bu/acre

Sep 1: 165.7 +/- 14.9 bu/acre

Oct 1: 170.9 +/- 10.3 bu/acre

Post Harvest: 174.3 +/- 7.0 bu/acre

Nebraska

Aug 1: 165.6 +/- 19.9 bu/acre

Sep 1: 174.6 +/- 15.7 bu/acre

Oct 1: 180.0+/- 10.8 bu/acre

Post Harvest: 183.6 +/- 7.3 bu/acre

Missouri

Aug 1: 137.0 +/- 16.4 bu/acre

Sep 1: 144.4 +/- 13.0 bu/acre

Oct 1: 148.9 +/- 8.9 bu/acre

Post Harvest: 151.9 +/- 6.1 bu/acre

Colorado

Aug 1: 116.0 +/- 13.9 bu/acre

Sep 1: 122.3 +/- 10.0 bu/acre

Oct 1: 126.1 +/- 7.6 bu/acre

Post Harvest: 128.6 +/- 5.1 bu/acre

TENACITY

- Jupyter notebook hosted by AWS and SageMaker crashed multiple times.
  - Work arounds included setting up a local environment in VS Code, however HLS data was inaccessible from local environments.

- Switched back to Jupyter after troubleshoot with AWS team in order to pipeline data into AWS-S3 bucket.

- Performed wifi speed ookla tests in multiple rooms to find optimal wifi connectivity points to work.

- Many issues loading the datasets due to not enabling exterior access to the jupyter labspace.
  - Fixed configuration settings to make sure it worked properly within the notebook.

Thank you!